

Skills Summary

Explaining Legal Moves in Equations and Inequalities

Use this reference while completing your worksheet or when studying.

Skill 1 — Naming the property behind a move

Think of a balance. Whatever you do to one pan you must do to the other, or it tips.

- Add the same amount to both sides — *addition property of equality*.
- Subtract the same amount from both sides — *subtraction property*.
- Multiply or divide both sides by the same non-zero number — *multiplication* or *division property*.
- Rewrite ONE side into an equal expression (distribute, collect like terms) — that is the distributive property or plain arithmetic, not a property of equality. It is still legal.

Example: Solve $2x - 9 = 5$ and name each move.

Step 1. The variable is tied up by a subtraction and then a multiplication, so I undo them in the reverse order, saying what each move is as I go.

Step 2. Add 9 to both sides (addition property): $2x = 14$.

Step 3. Divide both sides by 2 (division property): $x = 7$.

$$\Rightarrow x = 7$$

Try it: Solve $3x - 4 = 11$, naming each move as you go.

check: $5 = x$

Skill 2 — Solving when the variable is on both sides

Order of work: clear brackets and fractions first (one-side rewrites), then collect the variable on one side with a both-sides addition or subtraction, then finish with a both-sides division. Every line you write must be true whenever the line above it is true — that is the whole test of a legal move.

Example: Solve $5(x + 2) = 3x + 16$.

Step 1. There is a bracket on one side and an x on both, so I clear the bracket first and only then move the variable across.

Step 2. Distribute on the left (one-side rewrite): $5x + 10 = 3x + 16$.

Step 3. Subtract $3x$ from both sides: $2x + 10 = 16$. Subtract 10 from both sides: $2x = 6$.

Step 4. Divide both sides by 2.

$$\Rightarrow x = 3$$

Try it: Solve $4(x - 1) = 2x + 6$.

check: $5 = x$

Skill 3 — The one move that flips an inequality

Rule: adding or subtracting the same amount on both sides never changes the direction of an inequality. Multiplying or dividing both sides by a *negative* number does: the symbol must be reversed, because a negative multiplier swaps which side is larger. Multiplying by a positive number changes nothing. Always check one number from your answer set at the end.

Example: Solve $-2x + 1 \geq 9$.

Step 1. The variable has a negative coefficient, so I expect exactly one flip — at the division step, not before it.

Step 2. Subtract 1 from both sides (no flip): $-2x \geq 8$.

Step 3. Divide both sides by -2 , and reverse the symbol: $x \leq -4$. Check $x = -5$: $10 + 1 = 11 \geq 9$, true.

$$\Rightarrow x \leq -4$$

Try it: Solve $-4x \leq 24$.

check: $x = -6$